

香港中學文憑考試

Hong Kong Diploma of Secondary Education Examination

Economics

2020 Question Papers

(with marking schemes and comments on candidates' performance)



香港考試及評核局
Hong Kong
Examinations and
Assessment Authority

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Introduction

This publication provides a comprehensive report on the Economics examination in the 2020 Hong Kong Diploma of Secondary Education Examination. It covers various aspects of the examination, including the objectives, contents and format of the examination, the question papers, the marking schemes, and the chief examiners' comments on candidates' performance in the live examination. Some statistics are also included.

Through this publication, we hope to provide useful information and feedback to assist teachers in their teaching of the subject and students in their learning. Similar publications are also available for other subjects in the 2020 HKDSE. Besides, a series of publications and resource materials related to the HKDSE are also available for reference to schools, candidates and the public. These include information related to examination administration, assessment requirements, examination-related statistics and feedback provided to schools and other stakeholders. For details, please visit the Authority's website (http://www.hkeaa.edu.hk/en/HKDSE/info_corner/hkdse_publications_materials/).

We would like to take this opportunity to express our gratitude to all the examination personnel who were involved in the development of question papers and the assessment of candidates' performance. Without their contributions, it would not have been possible for us to conduct the examination smoothly.

The Examination

Assessment Objectives

The objectives of this assessment are to evaluate students' abilities and achievements in the following:

1. to know and understand fundamental economic concepts and theories;
2. to apply such concepts and theories to explain real world situations, especially the Hong Kong economy;
3. to understand and interpret economic information presented in various forms;
4. to acquire the basic tools of economic analysis;
5. to analyze economic issues;
6. to evaluate arguments, proposals and policies from different perspectives and make informed judgments; and
7. to present ideas clearly, and in a well-reasoned manner, illustrating answers with diagrams and examples.

Synopsis of curriculum contents

For details, please refer to the Economics Curriculum and Assessment Guide (Secondary 4 to 6) jointly prepared by the Curriculum Development Council and the Hong Kong Examinations and Assessment Authority.

Compulsory Part

1. Basic Economic Concepts
2. Firms and Production
3. Market and Price
4. Competition and Market Structure
5. Efficiency, Equity and the Role of Government
6. Measurement of Economic Performance
7. National Income Determination and Price Level
8. Money and Banking
9. Macroeconomic Problems and Policies
10. International Trade and Finance

Elective Part

Elective Part 1: Monopoly Pricing, Anti-competitive Behaviours and Competition Policy; OR
Elective Part 2: Extension of Trade Theory, Economic Growth and Development

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2020

ECONOMICS PAPER 1

8:30 am – 9:30 am (1 hour)

INSTRUCTIONS

1. Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the 'Time is up' announcement.
2. When told to open this book, you should check that all the questions are there. Look for the words '**END OF PAPER**' after the last question.
3. All questions carry equal marks.
4. **ANSWER ALL QUESTIONS.** You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.
5. You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.
6. No marks will be deducted for wrong answers.

Not to be taken away before the end of the examination session

There are 45 questions in this paper. Choose the **BEST** answer for each question.

1. If a command economy turns into a market-oriented economy,
 - (1) the income distribution of the economy will be more uneven.
 - (2) more resources will be allocated by price mechanism.
 - (3) people will have private property rights over all resources.
 - A. (1) only
 - B. (2) only
 - C. (3) only
 - D. (1), (2) and (3)

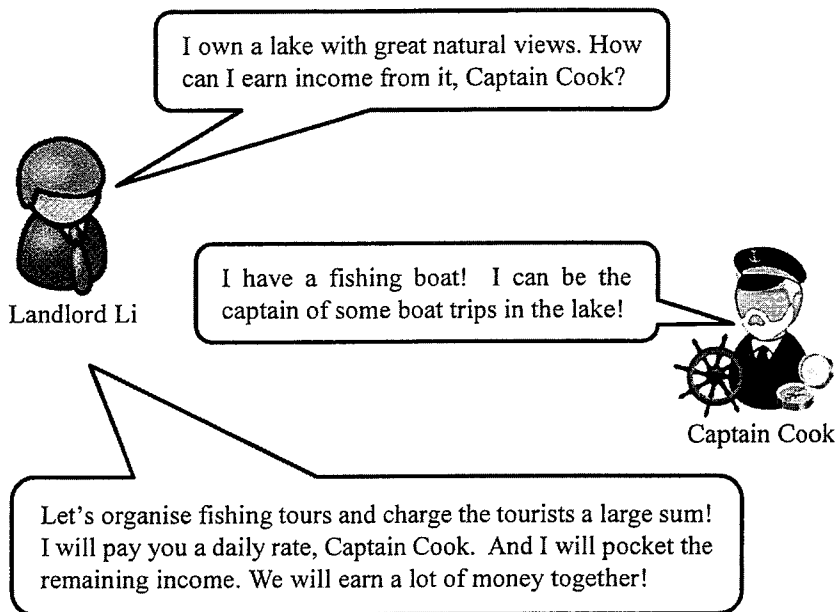
2. In Lunar New Year, tourists reserve tables in waterfront restaurants to enjoy firework display. The firework display is a _____ because _____.
 - A. public good the government pays for the firework display
 - B. public good one individual's consumption will not reduce its availability to others
 - C. private good private firms have to use resources to produce the firework display
 - D. private good the seats in waterfront restaurants are excludable in consumption

3. Which of the following may exist in a barter economy?
 - (1) specialisation
 - (2) inflation
 - (3) interest
 - A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

4. Some small investors prefer debentures to shares. Which of the following is a possible reason for that?
 - A. The return of debentures is generally higher than that of shares.
 - B. The price of shares may rise while that of debentures is fixed.
 - C. Shareholders need to spend time managing the company but debenture holders do not.
 - D. Companies have obligation to redeem debentures but they have no such obligation for shares.

5. If a public limited company in Hong Kong gets listed on the stock exchange,
- A. it will face a higher profits tax rate.
 - B. it can no longer keep its accounting information confidential.
 - C. its shareholders will enjoy higher liquidity of their assets.
 - D. it will enjoy economies of scale.

6. Refer to the following dialogue.



Given the return of land is rent, the return of labour is wage, the return of capital is interest while the return of entrepreneur is profit. The income earned by Captain Cook consists of _____ while the remaining income earned by Landlord Li consists of _____.

- A. wage.....profit
 - B. interestrent and profit
 - C. wage and interest.....rent
 - D. wage and interestrent and profit
7. Tai On Building (太安樓) in eastern Hong Kong Island is well-known as a 'foodie paradise'. A variety of street snacks such as cart noodles, egg waffles, dumplings and mango mochi are sold by numerous small vendors. Customers from different districts fill their stomachs at Tai On Building.

Based on the above information, street snack vendors at Tai On Building can _____.

- A. enjoy internal economies of scale
- B. enjoy external economies of scale
- C. suffer internal diseconomies of scale
- D. suffer external diseconomies of scale

8. The following table shows the production plan of a firm which employs two factors only, namely labour and machinery.

Machinery (units)	Labour (units)	Average product of labour (units)
4	1	20
4	2	21
4	3	22
4	4	X

If the above data illustrate the law of diminishing marginal returns, the value of X _____.

- A. can be 23
- B. can be smaller than 22
- C. must be larger than 22
- D. must be smaller than 22

9. Refer to the following table about the cost-output relationship of a profit-maximising firm.

Quantity (units)	1	2	3	4	5
Average cost (\$)	5	6	7	8	9

If the average cost of every unit of output decreases by \$1 and the market price is \$8,

- A. the minimum cost of production is \$18.
- B. the profit maximising output is 3 units.
- C. the maximum profit is \$10.
- D. the marginal cost of every unit of output will remain unchanged.

10. Alfred booked a room in a resort hotel. Later he found that Betty had booked a room in the same hotel at a lower price. Which of the following are possible reasons for such a price difference?

- (1) Alfred's booking was non-refundable while Betty's booking was with free cancellation.
- (2) Alfred booked the room in a weekend while Betty booked the room in a weekday.
- (3) Alfred booked the room with sea view while Betty booked the room with street view.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

11. When the price of Australia's red wine decreases from \$70 to \$50, the quantity demanded increases from 100 units to 122 units. What is the arc elasticity of demand for Australia's red wine within the above price range?

- A. 0.59
- B. 0.77
- C. 1.30
- D. 1.68

12. When the market supply curve of a good is derived, which of the following is/are assumed to be constant?

- (1) the market price of the good
- (2) the prices of factors of production of the good
- (3) the expected future price of the good

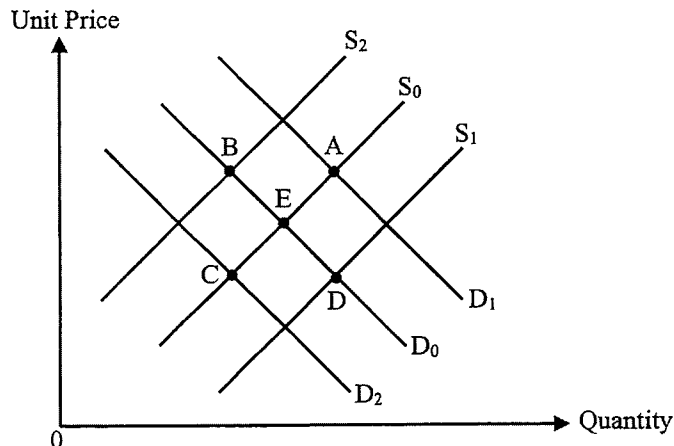
- A. (1) only
- B. (2) only
- C. (3) only
- D. (2) and (3) only

13. In recent years, the rise in demand for private cars in China has attracted Volkswagen, a German car manufacturer, to set up car manufacturing factories in China.

The above case illustrates the _____ function of price because _____.

- A. rationing private cars are sold to the highest-valued users in China
- B. rationing fewer buyers are willing to buy private cars in Germany
- C. allocative the supply of private cars falls in Germany
- D. allocative more resources are allocated to the production of private cars in China

14. Refer to the following supply-demand diagram for Good Y.



Suppose Good Y is an inferior good and the initial market equilibrium point is E. If there is a fall in the average income level of consumers, the new market equilibrium point will be _____.

- A. A
- B. B
- C. C
- D. D

15. The following table shows the demand and supply schedules of a good.

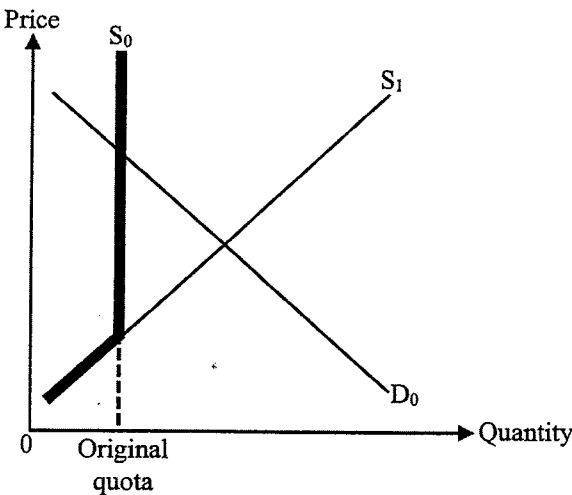
Price (\$)	Quantity demanded (units)	Quantity supplied (units)
20	70	40
22	60	45
24	50	50
26	40	55
28	30	60

If the government provides a \$6 per-unit subsidy for the producers of the good, the total subsidy provided by the government will be _____ and the ratio of consumers' share to producers' share of the subsidy will be _____.

- A. \$300 1:1
- B. \$300 1:2
- C. \$360 1:1
- D. \$360 1:2

16. Refer to the following supply-demand diagram for the exported Chinese rare earth minerals.

Exported rare earth minerals



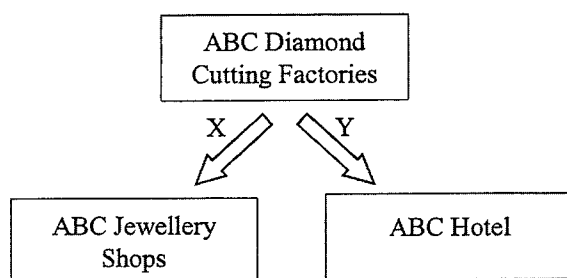
If China abolished the quota on the export of rare earth minerals,

- (1) the export price of the rare earth minerals would drop.
 - (2) the export quantity of the rare earth minerals would rise.
 - (3) the total exchange value of the exported rare earth minerals would drop.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

17. A non-profit organisation supports the increase in cigarette tax. The organisation reveals that every 10% increase in cigarette prices reduces the total cigarette quantity transacted by 4%. Which of the following statements about cigarette tax is **INCORRECT**?

- A. Law of demand is illustrated in the above case.
- B. The total expenditure on cigarette will increase after the increase of the cigarette tax.
- C. The increase in cigarette tax may lead to a greater income inequality.
- D. The cigarette tax could help reduce the problem of under-production of cigarette.

18. ABC Diamond Cutting Factories expand its business as follows:



To ABC Diamond Cutting Factories, the types of expansion shown by X and Y are _____ expansion and _____ expansion respectively.

- A. backward conglomerate
- B. backward lateral
- C. forward conglomerate
- D. forward lateral

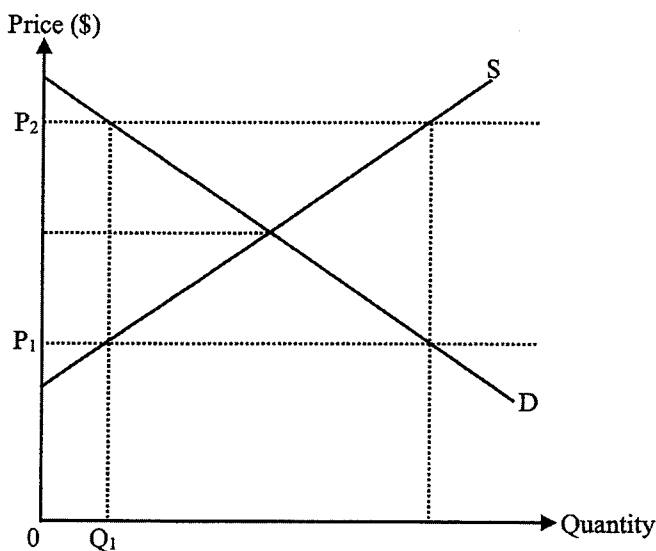
19. Study the following news extract.

In the last decade, the rent of residential units in Berlin was doubled due to continuous increase in demand. To deal with the problem, the Berlin government imposed a 'Rent Frozen Bill' under which the rent of residential units was not allowed to increase for at least 5 years.

Suppose a similar bill is implemented in Hong Kong. Which of the following statements is correct?

- A. The Gini coefficient will increase.
- B. The property tax revenue of the government will increase.
- C. There will be a deadweight loss if the demand for residential units keeps on increasing.
- D. The property owners will have higher incentives to renovate their flats.

20. Refer to the following supply-demand diagram for Good X.



If the price floor of Good X changes from P_1 to P_2 ,

- A. producer surplus will increase.
- B. quantity transacted of Good X will remain at Q_1 .
- C. deadweight loss will increase.
- D. excess demand of Good X will change to excess supply.

21.



The White Moose Café

May 5 at 1:31 AM · 🌟

****SCREAMING CHILD TAX****

For the benefit of both staff and customers alike, we have taken the decision to impose a tax on children who scream. If we notice that your child is screaming, an automatic 15% surcharge will be applied to your bill. For every further screaming child you have in your group, an additional 10% will be added.

We trust you understand our position.

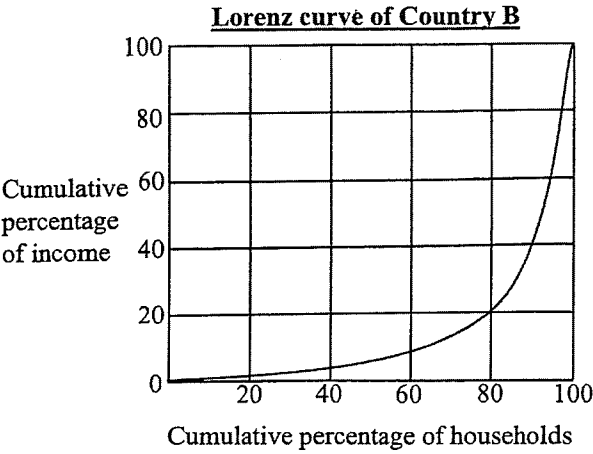
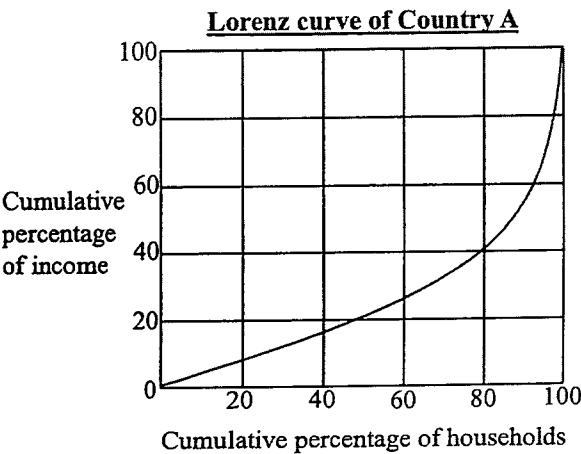
👍👍👍 6.2K

1.4K Comments 533 Shares

With reference to the above information, which of the following statements is correct?

- A. Children screaming in the café results in negative externality and all cafés should introduce similar policy to restore efficiency.
- B. The extra charge on screaming children increases the marginal private cost of the groups with screaming children.
- C. The 'screaming child tax' is a government solution to the divergence between private and social costs.
- D. The 'screaming child tax' is a kind of discrimination and should not be imposed.

22. Study the following two Lorenz curves.



Compared with Country B,

- A. the Gini coefficient of Country A is larger.
- B. Country A has a higher standard of living.
- C. the top 20% of households in Country A have less total income.
- D. Country A has a more even income distribution.

23. The following table shows the data of average annual income earned by women and men according to their age in 2018.

Age	Average annual income of women (HK\$)	Average annual income of men (HK\$)
20 - 29	122 000	121 000
50 - 59	482 000	645 000

Which of the following may explain the phenomena shown by the above data?

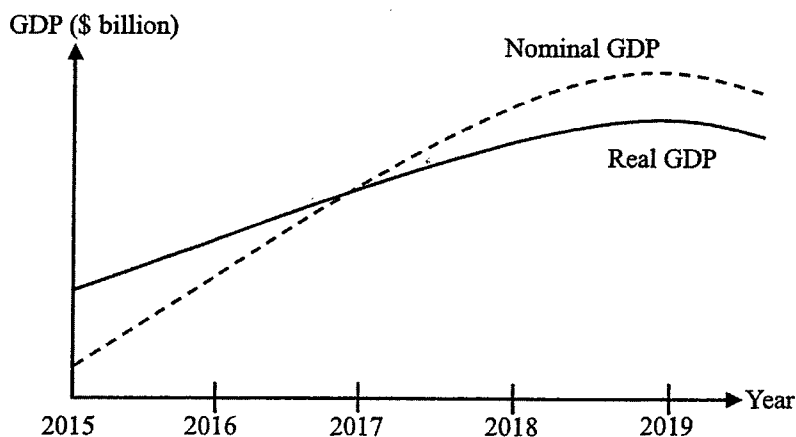
- (1) Women tend to leave the job market earlier than men for taking care of the family, leading to fewer women to take up senior positions in companies.
 - (2) Sex discrimination in labour market exists in the older generations.
 - (3) Accumulation of human capital contributes to the difference in income between different age groups.
- A. (1) only
 - B. (2) only
 - C. (3) only
 - D. (1), (2) and (3)

24. The following table shows the employment data of Hong Kong from April to June in 2019.

	Number of persons ('000)
Population	7 524
Aged 15 or above	6 590
Employed persons	3 871
Unemployed persons	114
Underemployed persons	41

According to the above data, the underemployment rate is _____.

- A. 0.55%
 B. 0.62%
 C. 1.03%
 D. 1.06%
25. Which of the following should be deducted from the gross domestic product of Hong Kong to calculate the gross national income of Hong Kong?
- A. the fee paid by a Hong Kong football club to a Japanese football team for an exhibition match in Hong Kong
 B. the rent earned by a Hong Kong property developer from its residential property in Japan
 C. the capital gain earned by a Japanese insurance corporation from its financial investment in Hong Kong
 D. the dividend paid by a listed company in Japan to a Hong Kong resident
26. The following diagram shows the gross domestic product (GDP) of Country A.



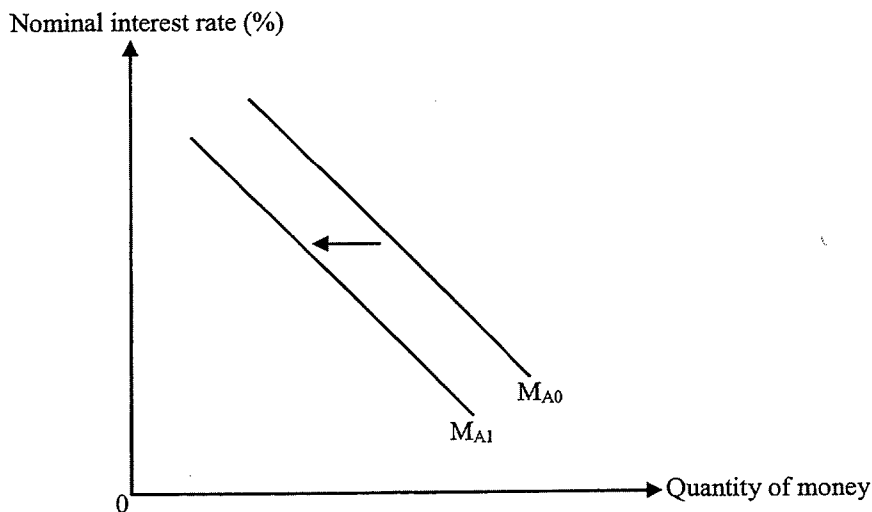
Which of the following statements about Country A is correct?

- A. The price level increased between 2016 and 2018.
 B. Deflation occurred before 2017.
 C. The actual inflation rate was equal to the expected inflation rate in 2017.
 D. The nominal GDP has been increasing since 2016.

27. Which of the following is a reason for a leftward shift of the short run aggregate supply curve in a country?

- A. There is a continuous increase in price level.
- B. The largest market of the country's exports suffers from an economic recession.
- C. More citizens emigrate to foreign countries.
- D. The wages of labour slowly adjust downward due to institutional factors.

28. The following diagram shows the asset demand for money of an economy.



Which of the following will cause the above change in asset demand for money?

- A. The nominal interest rate increases.
- B. There is a contraction of monetary base.
- C. The use of electronic wallets such as Alipay (支付寶) becomes more popular.
- D. The expected rate of return of holding bonds increases.

29. The legal tender in Hong Kong held by licensed banks is included in _____.

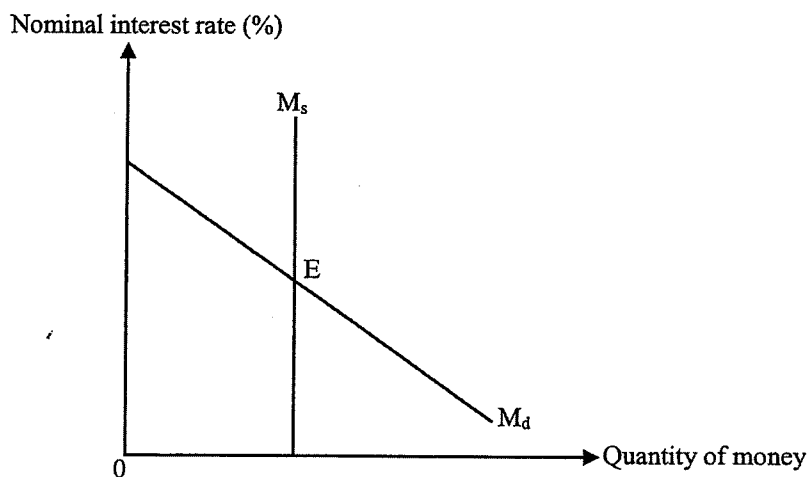
- (1) monetary base
 - (2) money supply M2
 - (3) money supply M3
- A. (1) only
 - B. (2) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

30. Refer to the following information about an economy. Suppose all banks do not hold any excess reserves.

	\$ million
Currency in public circulation	10
Monetary base	30
Money supply	110

If the public deposits all the cash into the banking system,

- A. the actual banking multiplier will be 4.
 - B. the maximum amount of loans will be \$120 million.
 - C. the maximum amount of deposits will be \$140 million.
 - D. the new money supply will be \$160 million.
31. The following diagram indicates the money supply curve (M_s) and money demand curve (M_d) of a certain economy. Suppose the initial equilibrium point is at E.

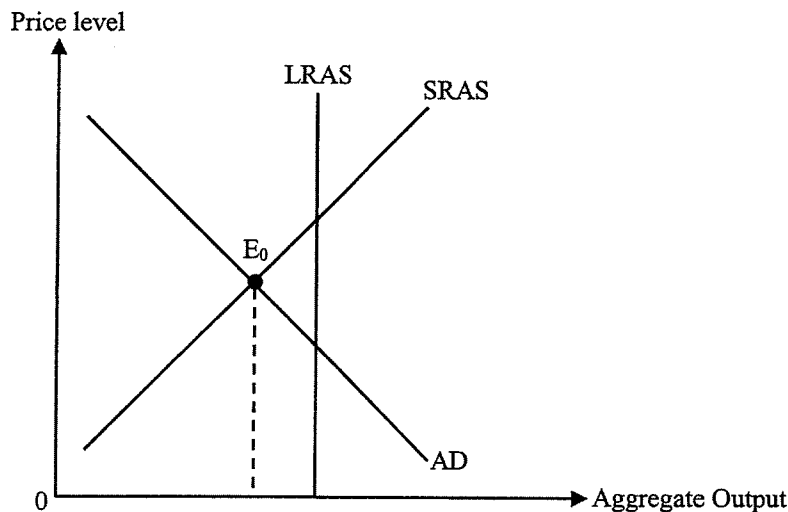


If the income of the public increases and the central bank sells government bonds to the public, the interest rate _____ and the quantity of money _____.

- A. will decrease will increase
- B. will increase may increase or decrease
- C. may increase or decrease will decrease
- D. will increase will decrease

32. Which of the following is a contractionary monetary policy?
- A. an increase in the discount rate
 - B. an increase in the salaries tax rate
 - C. a decrease in the required reserve ratio
 - D. a decrease in the quota for migration
33. Which of the following will increase the actual banking multiplier?
- A. The public tends to hold more cash.
 - B. The use of credit card becomes more popular.
 - C. The central bank increases the required reserve ratio.
 - D. The interest rate of bank deposit decreases.
34. When there is unexpected inflation, which of the following persons will gain?
- A. a landlord who adjusts his rental according to the price index
 - B. a creditor who has loaned out a sum of money at a floating interest rate
 - C. an employer who signs a fixed nominal wage contract with his employees
 - D. a retired civil servant who receives a fixed monthly pension
35. Which of the following policies would offset the effect of an increase in the willingness to invest on the price level?
- (1) an increase in profits tax rate
 - (2) an increase in transfer payment
 - (3) a decrease in tax allowance
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

36. The diagram below shows the aggregate demand curve, the short run aggregate supply curve and the long run aggregate supply curve of an economy.



Suppose the economy initially operates at E_0 and the total import value equals to the total export value. If the government decreases the amount of social security assistance for the elderly, in the short run, the deflationary (output) gap will become _____ and there will be a _____.

- A. narrower trade surplus
 - B. narrower trade deficit
 - C. wider trade surplus
 - D. wider trade deficit
37. An economist said, 'The continuous fall in stock prices in our country has already affected the consumption expenditure as well as the money demand. We should implement a policy to relieve the effects on the output level and the interest rate.' Which of the following policies can stabilise the output level and the interest rate in the country at the same time in the short run?
(Assume the interest rate is mainly affected by the asset demand for money in this case.)
- A. The government increases cash subsidy to the poor.
 - B. The government gives tariff exemption to the imports of goods.
 - C. The central bank instructs the commercial banks to tighten the requirement of personal loans.
 - D. The central bank buys government bonds from the public.
38. In 2019, the Japanese government increases the general sales tax rate on most of the goods and services from 8% to 10%. As a result,
- (1) income inequality will be higher.
 - (2) the fiscal deficit will be eliminated.
 - (3) the invisible trade deficit will decrease.
- A. (1) only
 - B. (2) only
 - C. (3) only
 - D. (1), (2) and (3)

39. The following is an extract of the mission of the Inland Revenue Department.

Our Mission

We are committed to

- collecting revenue efficiently and cost-effectively
- enabling staff to acquire the necessary knowledge, skills and attitude so that they can contribute their best to the achievement of our vision

The above mission is in line with the _____ principle of taxation put forward by Adam Smith.

- A. equality
- B. certainty
- C. convenience
- D. economy

40. Mr. Chan runs a factory in mainland China and earns a huge profit. Recently he has bought a flat and a second hand electric car in Hong Kong, and received dividends from his shares of HSBC.

According to the above information, which type(s) of tax does he need to pay to the Hong Kong government?

- (1) Property tax
 - (2) Stamp duty
 - (3) Profits tax
 - (4) First registration tax
- A. (2) only
 - B. (1) and (4) only
 - C. (2) and (3) only
 - D. (1), (3) and (4) only

41. In the case **WITHOUT** tax exemption, John needs to pay \$15 000 and Mary needs to pay \$35 000 as income tax.

Suppose the government proposes the following income tax exemption measures:

	Amount of tax exemption
Measure I	75% of income tax, maximum \$30 000
Measure II	100% of income tax, maximum \$20 000

John will prefer _____ while Mary will prefer _____.

- A. measure I measure I
- B. measure I measure II
- C. measure II measure I
- D. measure II measure II

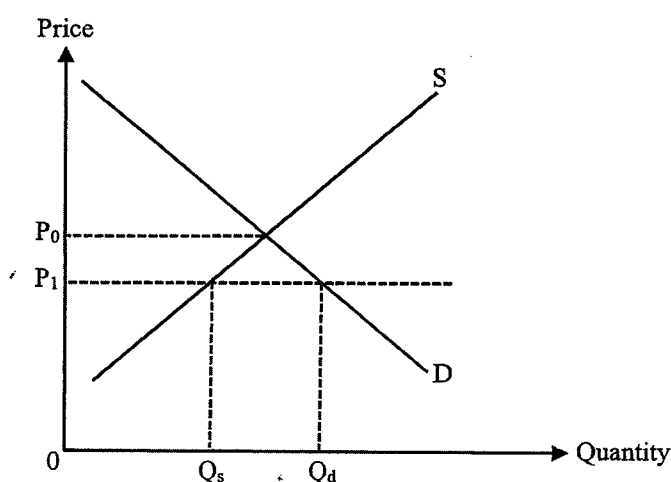
42. The following table shows the amounts of consumption on two goods in Country A and Country B before international trade. Assume both countries use one unit of resources on producing Good X and one unit of resources on producing Good Y.

	Consumption before trade (units)		
	Good X		Good Y
Country A	100	and	25
Country B	80	and	40

If Country A and Country B completely specialise their production and trade with each other, the total output of both countries will increase by _____ units of Good X and _____ units of Good Y.

- A. 15 15
- B. 15 20
- C. 20 15
- D. 20 20

43. The diagram below shows the market situation of Good X in a small open economy.



P_0 is the price before trade and P_1 is the price after trade. Which of the following statements about Good X are correct after the economy opens up for international trade?

- (1) The total expenditure on Good X of domestic consumers would be $P_1 \times Q_d$.
- (2) The value of domestically produced Good X would be $P_1 \times Q_s$.
- (3) The shortage of Good X would be $Q_d - Q_s$.

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

44. The following table shows the change in the exchange rate of Euro against the US dollar.

Date	Exchange rate
22 September 2018	1 EUR = US\$ 1.176
22 August 2019	1 EUR = US\$ 1.109

The above change in the exchange rate will lead to a decrease in

- A. the number of European tourists visiting the US.
 - B. the volume of goods exported from Europe to the US.
 - C. the total value of US exports to Europe (in terms of Euro).
 - D. the total value of European exports to the US (in term of the US dollar).
45. Under the linked exchange rate system of Hong Kong, which of the following statements is correct?
- A. The money supply M1 of Hong Kong has to be fully backed up by the US dollar.
 - B. Hong Kong's export volume will increase when the US dollar depreciates against other currencies.
 - C. The market exchange rate of the HK dollar relative to the US dollar is fixed.
 - D. Hong Kong citizens can sell 1 USD to the three note-issuing banks for 7.8 HKD.

END OF PAPER

Sources of materials used in this paper will be acknowledged in the *HKDSE Question Papers* booklet published by the Hong Kong Examinations and Assessment Authority at a later stage.

ECONOMICS PAPER 2
Question-Answer Book

10:15 am – 12:45 pm (2 hours 30 minutes)
This paper must be answered in English

INSTRUCTIONS

- (1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7, 9, 11 and 13.
- (2) This paper consists of three sections, A, B and C. Answer ALL questions in sections A and B. Attempt any ONE question in section C.
- (3) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (4) Supplementary answer sheets will be supplied on request. Write your candidate number, mark the question number box and stick a barcode label on each sheet, and fasten them with string INSIDE this book.
- (5) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

Please stick the barcode label here.

Candidate Number

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Section A (44 marks)

1. A free good must be free of charge. Do you agree? Explain your answer. (3 marks)

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2. A chain restaurant providing Chinese cuisine set up a cake shop.

(a) Explain which type of expansion this was. (2 marks)

(b) List **TWO** possible motives for the above expansion. (2 marks)

(c) After the opening of 10 more restaurants, the firm found that its average cost decreased. State **TWO** possible reasons for this. (2 marks)

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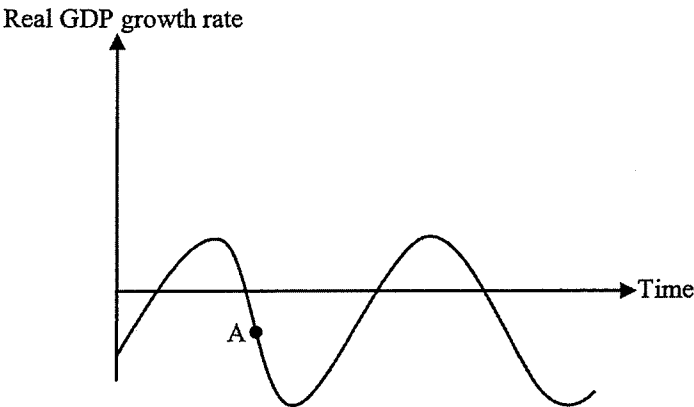
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Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

3. Below are the gross domestic product (GDP) data of a country.



- (a) Identify the phase of business cycle which point A is in. (1 mark)
- (b) Apart from increasing unemployment rate, list **TWO** general economic phenomena which occur at point A. (2 marks)
- (c) State **TWO** losses to society caused by unemployment. (2 marks)

Answers written in the margins will not be marked.

4. Study the following information of an economy.

Year	Consumer price index (CPI)
2018	110
2019	117

- (a)

Calculate the inflation rate in 2019.

(1 mark)
- (b)

Explain **ONE** difference between using CPI and GDP deflator to measure the change in the price level.

(2 marks)
- (c)

'The continuous open market purchase of government bonds by the central bank is the cause of inflation in the economy.' Justify the statement with reference to the classical quantity theory of money.

(4 marks)

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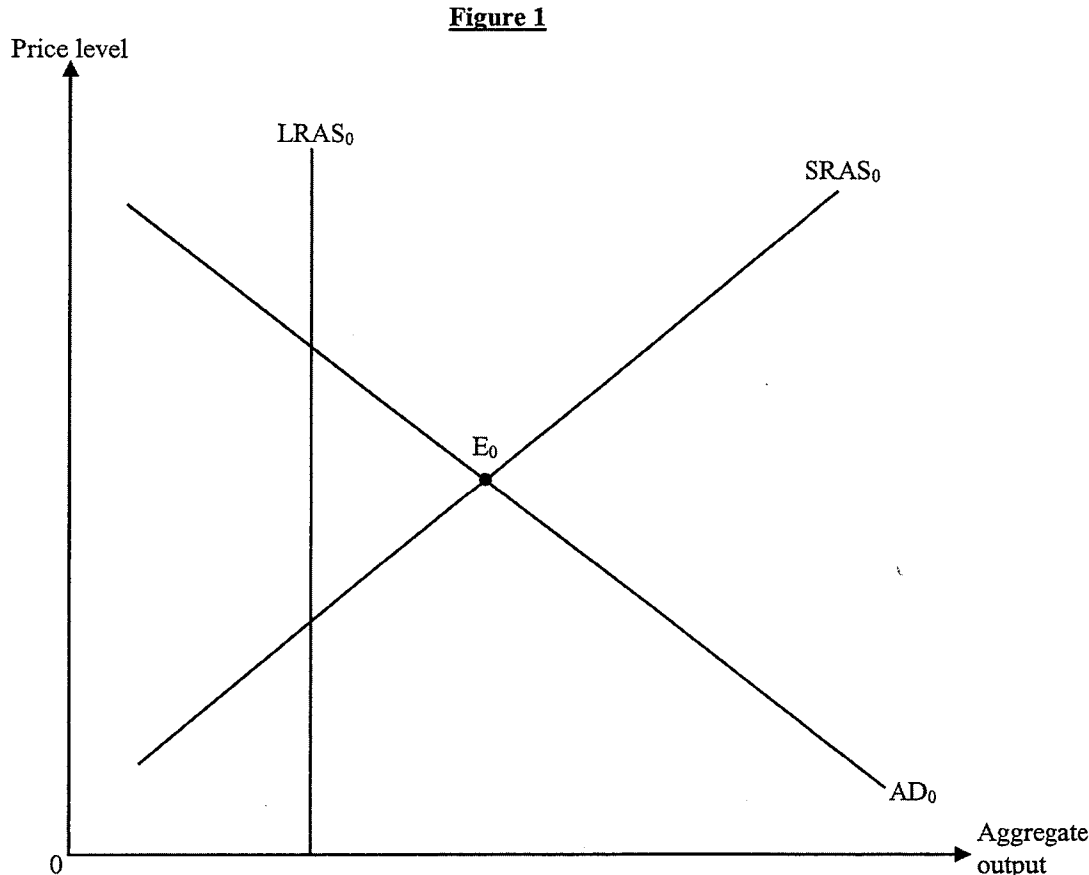
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5. (a) What is the opportunity cost of holding money? Explain. (2 marks)
- (b) Peter owes Mary \$100 and he has no cash in his wallet. He suggests repaying Mary by a cash coupon (with a face value of \$100) issued by a supermarket chain. Give **TWO** economic reasons to explain why Mary would refuse to accept the cash coupon as a repayment of Peter's debt. (4 marks)

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6. Refer to Figure 1.



Suppose an economy initially operates at E_0 . With the aid of Figure 1, explain how market forces can restore the aggregate output level to the full-employment output level in the long run. (5 marks)

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7. Suppose the outputs per unit of resources of Country A and Country B are as follows:

	Toys (units)		Cars (units)
Country A	10	OR	10
Country B	20	OR	10

- (a) Which good would Country B export according to its comparative advantage? Explain. (3 marks)
- (b) Suppose Country A and Country B trade at an exchange ratio of 5 units of toys to 4 units of cars. A transportation cost of 0.2 units of cars is required to exchange for 1 unit of toys and the cost is equally shared by both countries. Calculate the gain per unit of export of Country B. Show your workings. (3 marks)

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Answers written in the margins will not be marked.

Study the following balance sheet of the banking system in Country A.

Assets (\$million)		Liabilities (\$million)	
Reserves	1 000	Deposits	4 000
Loans	3 000		

Suppose the public always hold \$800 million cash and the banking system does not hold excess reserves.

- (a) Find the monetary base and money supply in Country A. (2 marks)
- (b) Suppose firms withdraw \$700 million from the banking system and remit the whole amount overseas.
 - (i) Find the change in monetary base in Country A. (1 mark)
 - (ii) Calculate the maximum possible change in money supply in Country A. Show your workings. (3 marks)

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END OF SECTION A

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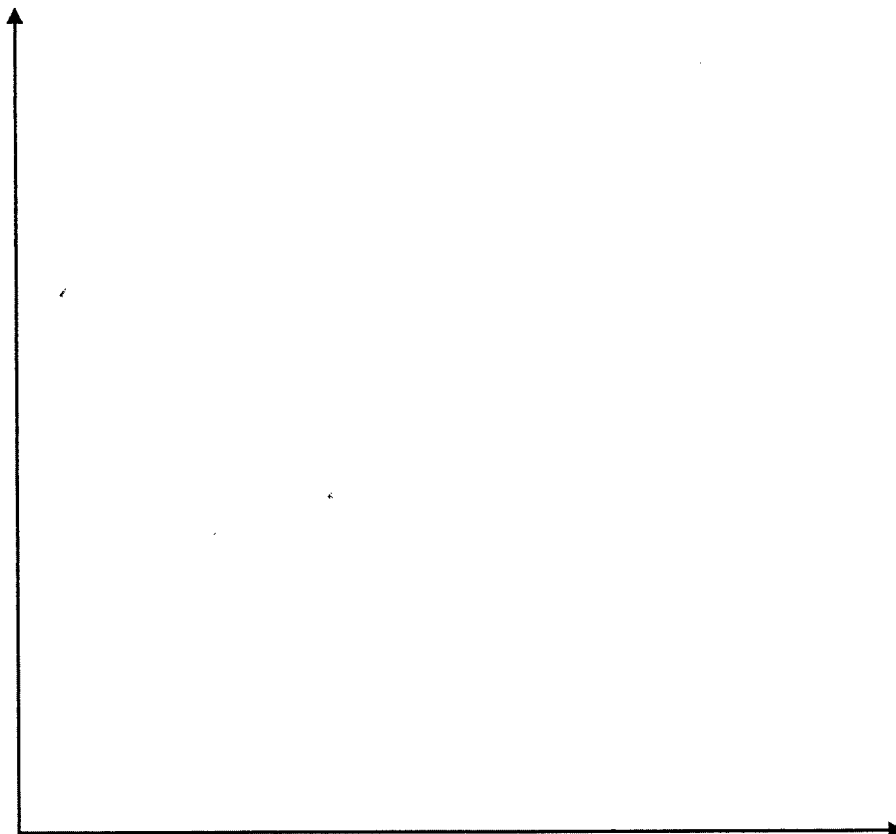
Section B (60 marks)

9. Ocean Park and Hong Kong Disneyland are the two large theme parks in Hong Kong.

- (a) What is the market structure of the theme park industry in Hong Kong? List **TWO** features of this market structure. (3 marks)

The first hotel in Ocean Park was launched in 2018 while the second hotel and the water park were scheduled to be opened before 2021. It is expected that such infrastructure development can attract more tourists to stay longer in Hong Kong.

- (b) Other things being equal, what will be the effects of the infrastructure development on Hong Kong's aggregate demand and long run aggregate supply in the future? How about on aggregate output? Explain your answers with the aid of a diagram. (8 marks)



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(c) Instead of obtaining bank loans to finance the infrastructure development, it is suggested that Ocean Park can be listed in the stock market and issue shares to raise capital. State **TWO** advantages of issuing shares over obtaining loans from banks for raising capital. (2 marks)

(d) The contribution of the infrastructure development of Ocean Park to Hong Kong's GDP in 2019 could be smaller than the total amount invested in the two hotels and the water park. Explain why the statement may be correct with **TWO** reasons. (4 marks)

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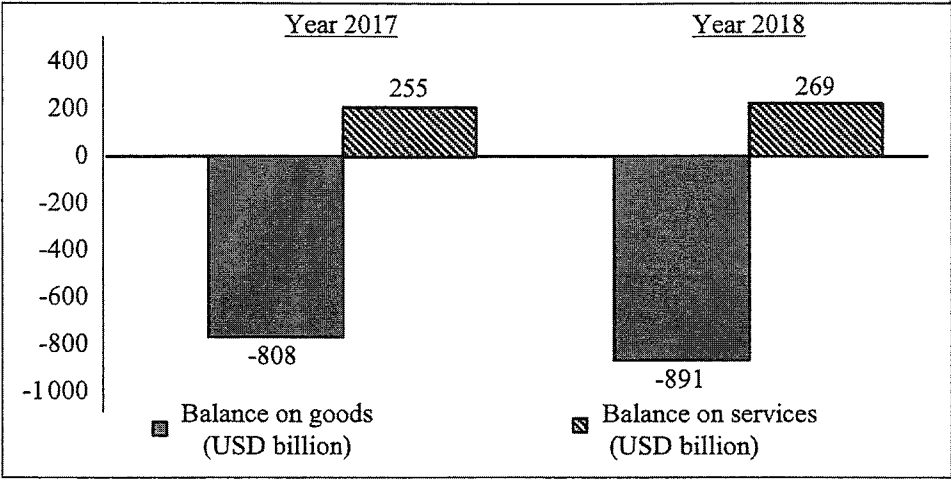
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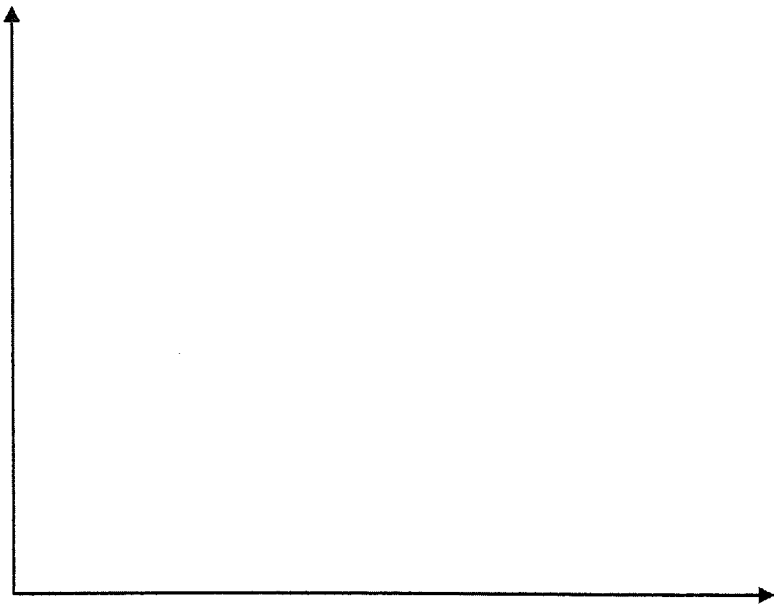
10. The following source shows the current account data of the US in 2017 and 2018.



- (a) (i) Describe the changes of current account data of the US in the above source. (2 marks)
- (ii) 'Based on the above source, we can conclude that the current account deficit of the US in 2018 *necessarily* increased.' Do you agree? Explain. (2 marks)

Donald Trump, the US president, once said that imposing tariffs would only increase the revenue of the US government and the US consumers did not have to pay more for the imported goods.

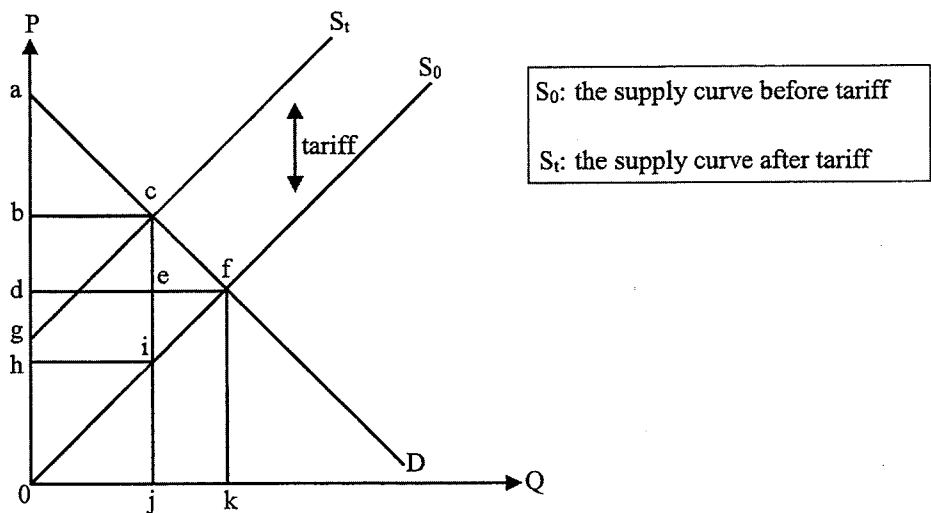
- (b) (i) With the aid of a supply-demand diagram, explain under **ONE** extreme condition that the consumers would bear **NO** tariff burden after the imposition of tariff. (4 marks)
(For simplicity, the tariff is assumed to be per-unit.)
- (ii) Indicate the tariff revenue (i.e. sellers' burden) in the same diagram. (1 mark)



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(c) Economists argue that tariff may harm both the consumers and producers.



With reference to the above supply-demand diagram, complete the table below. (3 marks)

	Before tariff	After tariff
Consumers' surplus	Area adf	
Producers' surplus		

(d) The US government could impose quota to restrict imports. From the viewpoint of the US government, state **ONE** advantage and **ONE** disadvantage of using tariff instead of quota to restrict imports. (2 marks)

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11. Traffic congestion in the cross harbour tunnels during peak hours is a serious problem in Hong Kong. For simplicity, the discussions below only focus on two cross harbour tunnels in Hong Kong, namely the Cross-Harbour Tunnel (CHT 紅隧) and the Western Harbour Crossing (WHC 西隧).

Source A: Existing tunnel tolls for private cars

CHT (owned by the government)	\$20
WHC (owned by a private firm)	\$75

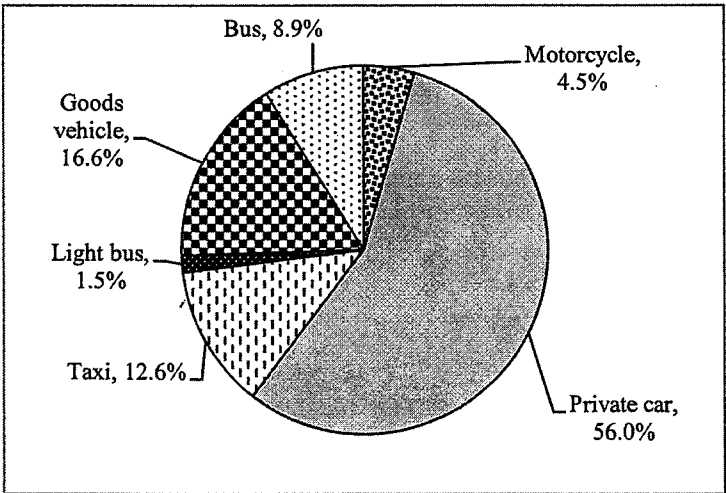
Source B: Traffic information during peak hours of the two tunnels

	Designed capacity per hour	Actual number per hour	Queue
CHT	2 600 (cars)	4 400-4 600 (cars)	1 500-3 000 (metres)
WHC	4 200 (cars)	3 600-3 800 (cars)	0 (metres)

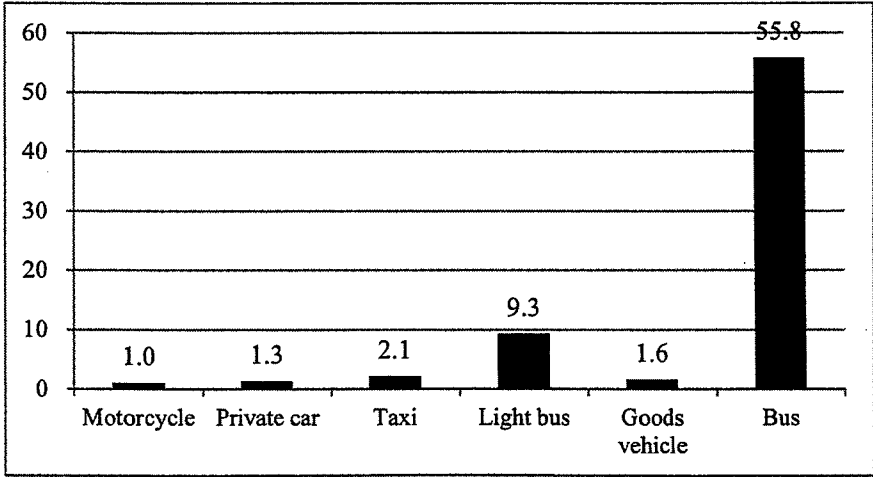
Source C: An extract of a report on traffic congestion

Traffic congestion not only affects individual drivers, but also results in inconvenience to society as a whole. Apart from the time wasted during congestion, people also need to allow extra time for the journey to cater for the uncertainty in travelling time. The stop-and-go traffic pattern during congestion increases the total operating costs of vehicles as more fuel would be consumed.

Source D: Percentage of vehicles crossing the harbour during peak hours (by types of vehicles)

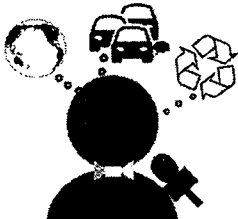


Source E: Average number of persons in each vehicle when crossing the harbour during peak hours



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Source F: Views on cross-harbour traffic in Hong Kong

A politician 	Every citizen needs to travel across the harbour. An increase in tunnel tolls affects everyone, especially the low-income group.
A government official 	We cannot underestimate the difficulty in negotiating any adjustment in tunnel toll of WHC. The government has no right to object to the tolls set by WHC.
An environmentalist 	Reducing traffic congestion in cross harbour tunnels is only one tiny step. The root of the problem is that there are too many cars in Hong Kong.
A taxi driver 	Channeling traffic to the under-utilised WHC benefits every road user and this allows taxi drivers to have more businesses.

Answers written in the margins will not be marked.

- (a) Refer to Source A. State **ONE** advantage and **ONE** disadvantage of providing cross harbour services by a private firm instead of the government. (2 marks)
- (b) Refer to Source B. With the aid of a supply-demand diagram, explain the phenomenon in the CHT (紅隧) during peak hours. (4 marks)
- (c) Refer to Source C.
- (i) Explain, with **ONE** example, why there may be a divergence between private and social costs during traffic congestion. (3 marks)
- (ii) Explain why the situation in part (i) may result in inefficiency. (2 marks)
- (d) According to Sources D and E, explain why private cars may be regarded as one of the major sources of traffic congestion in the cross harbour tunnels. (2 marks)
(Assume that the number of persons using the cross harbour tunnels per day during peak hours is constant.)

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- | | |
|-------------|--|
| Proposal I | The government raises the toll of the CHT (紅隧) for private cars to \$75. |
| Proposal II | The toll of the WHC (西隧) for private cars is reduced to \$20. |

- the owners of the WHC company
- the politicians who are concerned about the income inequality in Hong Kong
- the government economists who are concerned about government revenue and/or expenditure
- **ONE** other group of stakeholders

(14 marks)

- (2 marks)

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END OF SECTION B

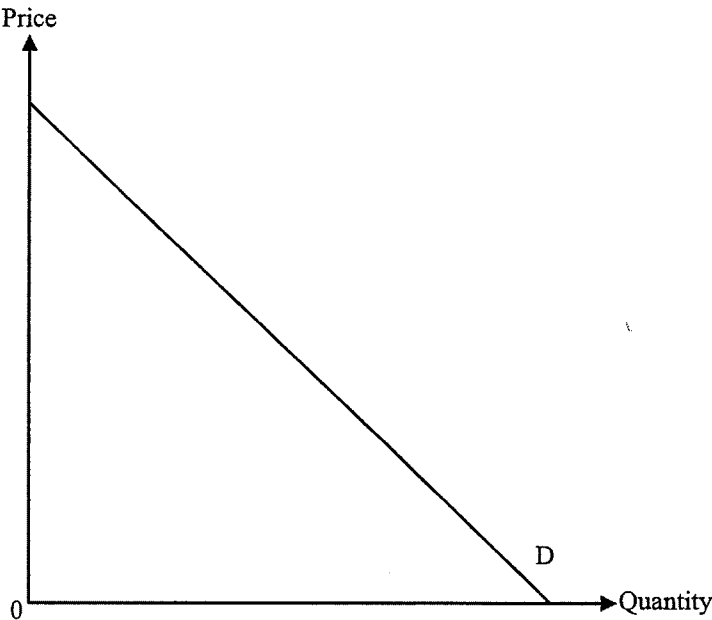
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Section C (16 marks) Answer any ONE question.

12. ABC Electricity Limited, the sole supplier of electricity in Country A, adopts uniform pricing.

- (a) If the marginal cost is constant, explain how the profit-maximising output of ABC Electricity Limited will be determined. Indicate the output (Q_M), the price (P_M) and the deadweight loss in Figure 2. (5 marks)

Figure 2



- (b) Suppose the government imposes a lump sum tax, which is a fixed amount regardless of the quantity produced, on ABC Electricity Limited. (Assume that the tax amount is smaller than the profit.)
- (i) Indicate the after tax profit-maximising output (Q_t) and price (P_t) in Figure 2. (2 marks)
- (ii) Explain whether economic efficiency would worsen as a result of the tax. (2 marks)
- (c) ABC Electricity Limited charges users in the urban area \$1 per unit of power consumed and users in the rural area \$1.5 per unit of power consumed. Define price discrimination and explain why this price arrangement may **NOT** constitute price discrimination. (3 marks)

ABC Electricity Limited planned to install new equipment for electricity supply in the rural area and asked the equipment providers to bid for the contracts. It was found that the two equipment providers in a region agreed to take turns being the bid winner.

- (d) (i) Which anti-competitive behaviour would the above case constitute? If the above case happened in Hong Kong, which conduct rule of Competition Ordinance would be violated? (2 marks)
- (ii) State **TWO** ways the above behaviour would harm ABC Electricity Limited. (2 marks)

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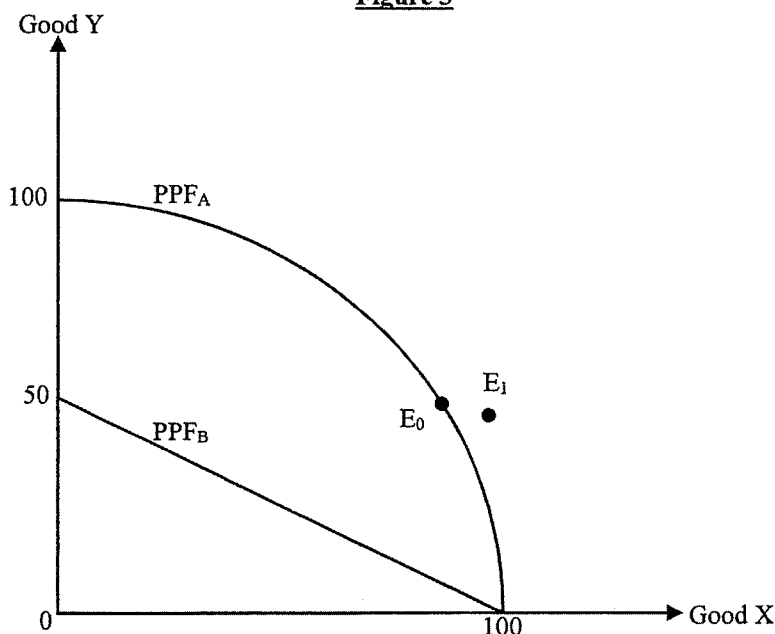
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13. Refer to Figure 3. PPF_A and PPF_B are the production possibility frontiers of Country A and Country B respectively. E_0 is the consumption point of Country A without trade.

Figure 3



Assume the domestic cost of production of 1 unit of Good X in Country A is 1.5 units of Y at E_0 .

- (a) With reference to the domestic production costs of Country A and Country B, explain which country will specialise in producing Good X. (3 marks)

- (b) After Country A engages in trade with Country B, the consumption point of Country A is E_1 .

Indicate in Figure 3,

- the consumption possibility frontier (CPF_A) of Country A (Hint: CPF_A should pass through E_1),
 - the production point (A_T) of Country A after trade and explain your answer in terms of terms of trade, and
 - the consumption possibility frontier (CPF_B) and the production point (B_T) of Country B after trade. (5 marks)
- (c) Suppose Country B adopts artificial intelligence technologies in the production of Good Y, which doubles her productivity of the good.
- Explain whether your answer in part (a) would be affected by the above change. (2 marks)
 - Give **TWO** reasons to explain why the adoption of artificial intelligence technologies may boost economic development of Country B. (4 marks)
 - Apart from adopting artificial intelligence technologies, list **TWO** factors that may boost economic growth. (2 marks)

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END OF PAPER

Sources of materials used in this paper will be acknowledged in the *HKDSE Question Papers* booklet published by the Hong Kong Examinations and Assessment Authority at a later stage.

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Supplementary Answer Sheet

Question No. _____

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Marking Schemes

Paper 1

Question No.	Key	Question No.	Key
1.	B (77%)	26.	A (43%)
2.	B (80%)	27.	C (53%)
3.	B (49%)	28.	D (51%)
4.	D (64%)	29.	A (31%)
5.	C (40%)	30.	B (70%)
6.	D (42%)	31.	D (52%)
7.	B (79%)	32.	A (66%)
8.	B (58%)	33.	B (36%)
9.	B (68%)	34.	C (73%)
10.	C (85%)	35.	B (71%)
11.	A (34%)	36.	C (56%)
12.	D (39%)	37.	D (51%)
13.	D (79%)	38.	A (68%)
14.	A (62%)	39.	D (67%)
15.	D (63%)	40.	A (54%)
16.	A (78%)	41.	C (57%)
17.	D (62%)	42.	C (71%)
18.	C (87%)	43.	A (46%)
19.	C (77%)	44.	A (70%)
20.	C (60%)	45.	B (69%)
21.	B (60%)		
22.	D (87%)		
23.	D (48%)		
24.	C (68%)		
25.	A (53%)		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

Paper 2

These documents were prepared for markers' reference. They should not be regarded as sets of model answers. Candidates and teachers who were not involved in the marking process are advised to interpret their contents with care.

The answers provided in the marking scheme are for reference only. They are not the only possible answers. Alternative answers are acceptable so long as they are well reasoned.

The examination emphasises the testing of the understanding of economic theories and the application of the knowledge of economic analysis to practical problems. Candidates are advised to study this document in conjunction with the examiner's comments on candidates' performance in this booklet.

For essay-type questions, candidates are expected to demonstrate an understanding of the question, an ability to deploy relevant knowledge of the subject in response to the questions, and to present their answers logically and coherently.

In questions asking for a specified number of reasons or examples etc. and a candidate gives more than the required number, the extra answers should not be marked. For instance, in a question asking candidates to provide two examples, and if a candidate gives three answers, only the first two should be marked.

The following symbols are used:

- / A single slash indicates an acceptable alternative within an answer.
- @ The number in front of the symbol indicates the marks for each point.
- max Maximum mark for the question/sub-question

Section A

- | | Marks |
|---|----------------|
| 1. Yes.
More of free good is not preferred. No one is willing to pay a price/give up resources for getting more free good. | (1)
(2) |
| 2. (a) Lateral expansion
because the firm expanded its production to related but not competitive products. | (2) |
| (b) <u>Motives:</u> <ul style="list-style-type: none">- market/product diversification- economies of scale- use of brand name- any other relevant point [Mark the FIRST TWO points only.] | } @1
Max: 2 |
| (c) <u>Reasons:</u> <ul style="list-style-type: none">- share of advertising cost by a larger number of output- bulk purchase of raw materials in production- more extensive division of labour- any other relevant point [Mark the FIRST TWO points only.] | } @1
Max: 2 |
| 3. (a) Recession. | (1) |
| (b) <u>Phenomena:</u> <ul style="list-style-type: none">- falling/negative real GDP growth rate OR falling real GDP level- falling inflation rate OR deflation- falling consumption/investment- falling business confidence- any other relevant point [Mark the FIRST TWO points only.] | } @1
Max: 2 |
| (c) <u>Losses caused by unemployment:</u> <ul style="list-style-type: none">- loss of output- loss of human capital- political and social unrest- any other relevant point [Mark the FIRST TWO points only.] | } @1
Max: 2 |

- | | Marks |
|--|---|
| 4. (a) Inflation rate = $(117 - 110) / 110 \times 100\% = 6.36\%$ | (1) |
| (b) <u>Differences:</u> | |
| <ul style="list-style-type: none"> – CPI only includes consumer goods while GDP deflator includes both consumer goods and producer goods. – The computation of CPI is based on a fixed basket of goods (with weights which would only be revised infrequently), while GDP deflator is based on variable baskets of goods (with weights varying from year to year). – any other relevant point <p>[Mark the FIRST point only.]</p> | <div style="font-size: 3em; line-height: 1; padding: 0 10px;">}</div> <div style="display: inline-block; vertical-align: middle;"> <p>@2</p> <p>Max: 2</p> </div> |
| (c) $MV = PY$, where M: money supply, V: velocity of circulation of money, P: price level and Y: output | |
| Assume V and Y being constant. | |
| A continuous increase in M (caused by the open market purchase) would result in a continuous increase in P, i.e. inflation. | (2) |
| (2) | |
| 5. (a) Nominal interest (rate) | |
| because, while holding money, one has to give up the opportunity to earn interest income by depositing the sum into banks or by using the sum to purchase interest-earning assets like bonds. | (1) |
| (1) | |
| (b) <u>Reasons:</u> | |
| <ul style="list-style-type: none"> – Not generally accepted as a medium of exchange, as the coupon can only be used in the supermarket chain but not in other shops. – Not divisible as the coupon, is difficult to be broken down into smaller units to facilitate exchange of cheaper goods. – Not a good store of value, as the coupon becomes worthless in case the supermarket chain closed down. – Not durable as banknotes, because the coupon is not waterproof and may be easily damaged. – Not a legal tender, as other people may refuse to accept the coupon as payment. – any other relevant point <p>[Mark the FIRST TWO points only.]</p> | <div style="font-size: 3em; line-height: 1; padding: 0 10px;">}</div> <div style="display: inline-block; vertical-align: middle;"> <p>@2</p> <p>Max: 4</p> </div> |

Marks

6. Verbal elaboration:

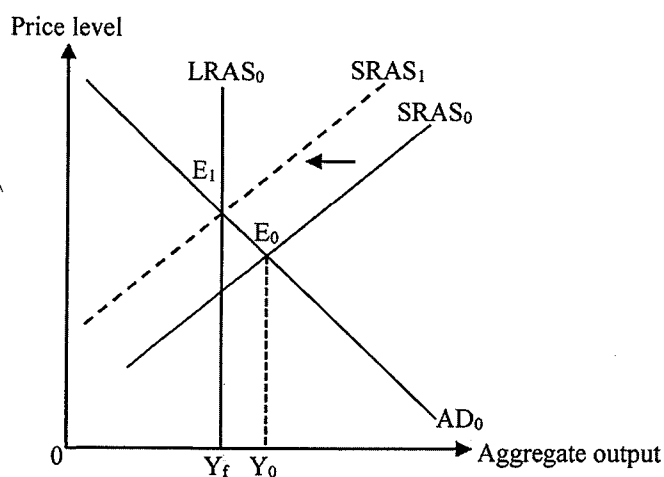
The economy is facing an inflationary (output) gap. There would be an excess demand in the factor market, creating pressure for factor prices to adjust upward. As the costs of production would rise, short run aggregate supply (SRAS) would fall over time, thus restoring output to Y_f in the long run.

(4)

Illustrate in the diagram:

– SRAS shifts leftward

(1)



7. (a) In country A, opportunity cost of producing 1T = 1C (3)
 In country B, opportunity cost of producing 1T = 0.5C
 Country B has a lower opportunity cost in producing T and would therefore export T.

- (b) Terms of Trade: 1T:0.8C (3)
 Gain from trade = $0.8C - 0.5C - 0.2C/2 = 0.2C$

8. (a) Monetary base: \$1000 million + \$800 million = \$1800 million (1)
 Money supply: \$800 million + \$4000 million = \$4800 million (1)

- (b) (i) Change in monetary base = -\$700 million (1)

- (ii) Change in money supply = $(\$300 \text{ million} \times \frac{1}{0.25} + \$800 \text{ million}) - \$4800 \text{ million}$ (3)
 = -\$2800 million

Section B

9. (a) Oligopoly.

Marks
(1)

Features:

- a few dominant sellers
- interdependency of pricing
- price and non-price competitions
- any other relevant point

[Mark the FIRST TWO points only.]

} @1
Max: 2

(b) Verbal elaboration:

When more tourists stay longer in Hong Kong, the tourist expenditure would increase, implying a rise in export of service. This would lead to an increase in AD.

(5)

On the other hand, the infrastructure development would expand the production capacity of the economy in the future, leading to an increase in (future) LRAS.¹

In the long run, aggregate output increases.

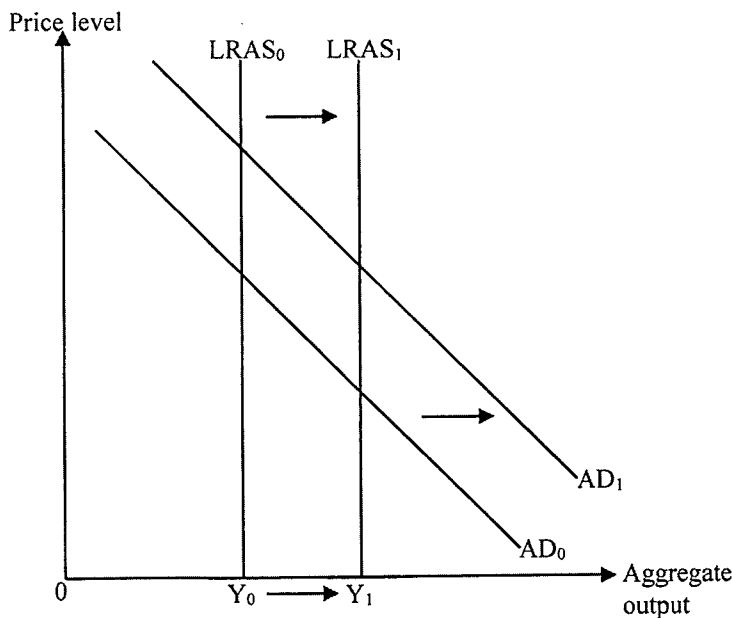
Illustrate in the diagram:

- AD shifts to the right
- LRAS shifts to the right
- $Y_1 > Y_0$

(1)

(1)

(1)



¹ One can also argue that the new hotels and the waterpark may not be able to cause a significant increase in long run aggregate supply, as such development has direct effect only on one single industry, which may be of minor importance relative to the production capacity of the economy as a whole.

	Marks
(c) <u>Advantages:</u>	
– no interest obligation	(1)
– no repayment obligation ²	(1)
– any other relevant point	
[Mark the FIRST TWO points only.]	
(d) <u>Reasons:</u>	
– Not all factor inputs used to build the infrastructure originated from Hong Kong. For example, some raw materials were imported.	(2)
– Some of the output might not be produced in 2019. For example, some has been already included as change in inventory/work in progress in previous years.	(2)
– any other relevant point	
[Mark the FIRST TWO points only.]	
10. (a) (i) Trade deficit increased while invisible trade surplus slightly increased.	(2)
(ii) No. We also need the information of external factor income flow (i.e. net income from abroad) and current transfers to draw the conclusion.	(2)

² Under the current arrangement as specified by the Ocean Park Ordinance (Cap. 388), the Ocean Park as a public corporation is not allowed to issue shares to raise capital. But such knowledge is not required to answer this hypothetical question.

Marks

(b) (i) **Condition I: Ed infinite**

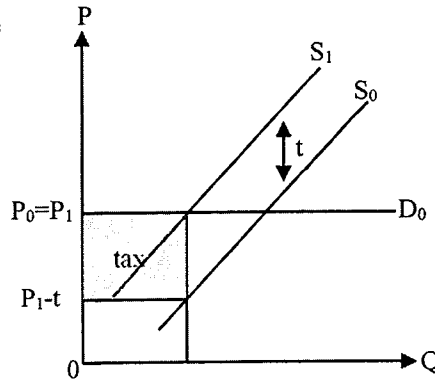
Verbal elaboration:

When demand is perfectly elastic, the market price would not increase even after the imposition of tariff. Since a tiny increase in price would drop the quantity demanded to zero, the supplier cannot shift the tax burden to the consumers at all.

(2)

Illustrate in the diagram:

- horizontal demand curve
- upward shifting of supply curve



(1)

(1)

OR

Condition II: Es zero

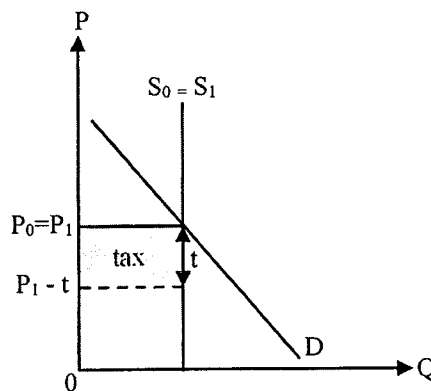
Verbal elaboration:

When supply is perfectly inelastic, the imposition of tariff would not cause the market price to increase. Given the quantity supplied is fixed, any increase in price would only result in an excess supply.

(2)

Illustrate in the diagram:

- vertical supply curves (old and new)
- indication of per unit tax



(1)

(1)

(ii) - correct amount of tax revenue indicated in the corresponding condition.

(1)

(c)

	Before tariff	After tariff
Consumers' surplus	-	Area abc
Producers' surplus	Area dfo	Area hi0

Marks
(3)

(d) Advantage:

- The government can earn the tariff revenue (while the government may not receive the quota rent).

(1)

Disadvantage:

- The effect of tariff on import volume depends more on elasticity (while the effect of quota on import volume is certain).

(1)

[Mark the FIRST advantage and FIRST disadvantage only.]

Marks

11. (a) Advantages:

- more efficient management
- easier to recruit experts
- any other relevant point

} @1
Max: 1

Disadvantages:

- higher toll as private firm aims at profit-maximisation
- less information for future planning
- any other relevant point

} @1
Max: 1

[Mark the FIRST advantage and FIRST disadvantage only.]

(b) Verbal elaboration:

When the price is set below the equilibrium (i.e., market-clearing) level, there is an excess demand.

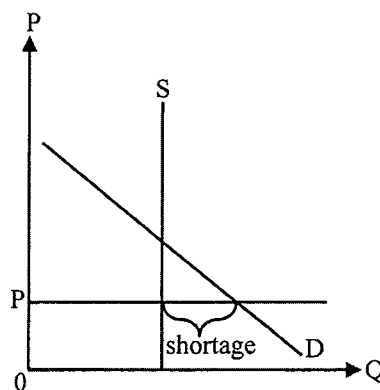
(2)

Illustrate in the diagram:

- vertical supply curve
- correct position of shortage

(1)

(1)



- Marks**
- (c) (i) The drivers using the tunnel would have adverse effects on other drivers without paying compensation to the parties being affected. These adverse effects can be viewed as an external cost, which the drivers would not consider when making their driving decisions.
The example of external cost includes extra time for journey to cater for uncertainty/extra operating cost in the stop-and-go traffic.

(ii) The above case implies that marginal social cost is larger than marginal social benefit. The tunnel is 'over-used'; the quantity of cars using the tunnel exceed its efficient level.

(d) Private cars consist of more than half of the tunnel traffic (from Source D) and the average number of persons in the vehicles is very low (from Source E).

(3)

(2)

(2)

(e) Marks award for effective communication (EC: max 2 marks)

Marks	Performance
2	● <i>Support arguments with the source/data and appropriate economic theories.</i> ● <i>Present relevant material.</i> ● <i>Present well-organised and coherent answers without repetition of ideas.</i> ● <i>Use language that expresses ideas clearly and fluently with appropriate use of words/terms/symbols.</i>
1	● <i>Present arguments with some support of the source/data and economic theories.</i> ● <i>Present some irrelevant material.</i> ● <i>Present answers in a less organised way with some repetition.</i> ● <i>Use language that conveys a clear message with some inappropriate use of words/terms/symbols.</i>
0	● <i>Present arguments with no support of the source/data and economic theories.</i> ● <i>Present material unrelated to the gist of the question.</i> ● <i>Present inconsistent arguments.</i> ● <i>Express limited ideas with inappropriate use of words/terms/symbols.</i>

The maximum marks for content is 12 marks

Answers may include the following:

WHC company:

CHT and WHC are substitutes. A rise in toll of CHT would increase the demand of WHC. With the same toll and higher quantity, the total revenue of WHC must rise. WHC company would prefer Proposal I.

WHC is a private firm who would be profit-maximising when it freely sets its price at \$75. Charging less would yield less profit (unless the government pays a subsidy for it). Therefore, WHC company would NOT prefer Proposal II.

Income inequality:

'Toll-as-regressive-tax' argument: tunnel toll is regressive in nature - although both the rich and the poor pay the same amount of toll, the toll consists of a smaller percentage of the income of the rich than that of the poor. Since an increase in such a regressive 'tax' would worsen the income inequality and Proposal I is NOT preferred, while a decrease of such a regressive 'tax' would improve the income inequality and Proposal II would be preferred.

OR

'Rich-driving-private-cars' argument: Mainly the rich travels with private cars while the poor mainly uses public transport to cross the harbour. So the adjustment of tolls on private cars mainly affects the rich but only have little effect on the poor. It implies that increasing the toll would lead to a more even distribution of income and Proposal I would be preferred.

Government revenue and / or expenditure:

Given the situation with excess demand, the increase in toll for CHT may only reduce the amount of excess demand without changing the quantity transacted, resulting in a higher total revenue. Therefore, Proposal I would be preferred.

If the demand of CHT is inelastic (due to its superior location/lack of substitutes), the increase in toll would lead to an increase in total revenue. Therefore, Proposal I would be preferred.

In order to induce WHC to cut its toll, the government may need to compensate the company for its losses by providing subsidies or tax benefits. As a result, such transfers from the government to WHC would reduce the net revenue of the government.³ Therefore, Proposal II would NOT be preferred.

³ Strictly speaking, such transfers should be treated as negative taxation rather than expenditure.

Taxi drivers:

Since private car trips and taxi trips are substitutes, the increase in toll of private car may lead to an increase in demand for cross-harbour-taxi-trip. As some private car drivers are priced out by the higher toll and there would be fewer vehicles in the tunnels. The overall improvement in traffic condition will induce a higher demand for taxi trips (as trips are faster on average) and the cost of operation will be lowered because of less stop-and-go traffic pattern will be observed. Taxi-drivers would therefore prefer Proposal I.

Environmentalists:

Some private car drivers are priced out by the higher toll and they may switch to public transport which implies more efficient use of tunnels and roads in general. Less congestion also implies less stop-and-go traffic and less air pollution. Environmentalists would thus prefer Proposal I.

(f) Proposal:

- Building a new tunnel: This would solve the overall shortage of cross-harbour capacity.
- Subsidising the use of public transport (MTR/bus/ferry): This may discourage the use of private cars, reducing the number of cars on road.
- Heavier first registration fee on private cars: This may reduce the number of cars on road.

Marks

} @2
Max: 2

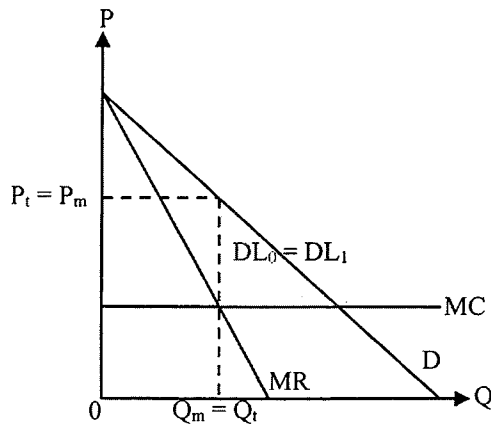
Section C

Marks

12. (a) Verbal elaboration:
The profit-maximising output is determined by the condition: marginal revenue (MR) = marginal cost (MC). (1)

Illustrate in the diagram:

- horizontal MC (1)
- correct position of MR (1)
- correct position of Q_m and P_m (1)
- correct position of DL (1)



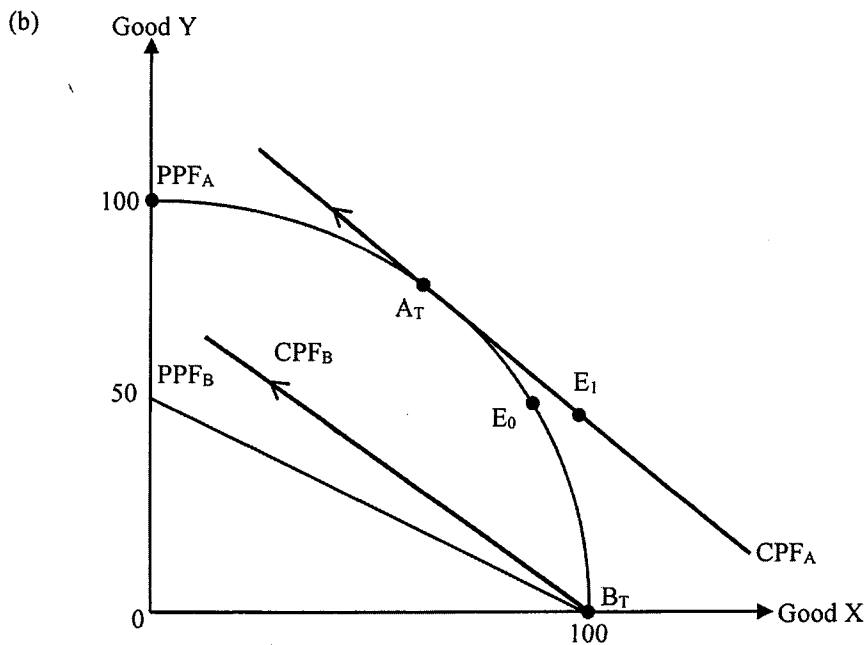
- (b) (i) Illustrate in the diagram:
- correct position of $P_t (=P_m)$ (1)
 - correct position of $Q_t (=Q_m)$ (1)
- (ii) No, since imposition of the lump-sum tax would not affect either demand or marginal cost, output and deadweight loss would remain unchanged. (2)
- (c) Price discrimination is defined as charging different consumers different prices for the same good produced with same costs. In this case, the production cost may not be the same. The installation cost and maintenance cost for rural area might be higher so it may lead to a higher price. (3)
- (d) (i) Bid-rigging. (2)
It violated the first conduct rule.
- (ii) Harm:
- higher price/cost
 - lower quality
 - fewer choices
 - any other relevant point
- [Mark the FIRST TWO points only.]

} 1@
max: 2

Marks
(3)

13. (a) The opportunity cost producing 1X in Country A = $1.5Y$
The opportunity cost of producing 1X in Country B = $\frac{50}{100} = 0.5Y$

Since Country B has a lower opportunity cost of producing good X, it would specialise in producing good X.



Illustrate in the diagram:

- correct position of CPF_A (1)
- correct position of A_T (1)
- correct position of CPF_B (1)
- correct position of B_T (1)

Verbal elaboration:

Terms of trade would be equal to the slope of PPF_A at point A_T (which is the same as the slopes of CPF_A and CPF_B). (1)

Marks
(2)

- (c) (i) No. The opportunity cost of producing 1X in Country B would become 1Y, which is still lower than that of Country A.

(ii) Reasons:

- Artificial intelligence technology could increase the per capita GDP as labour becomes more productive with better technology.
- Artificial intelligence technology may improve medicine production and surgery procedures and improve citizens' average life expectancy.
- Artificial intelligence technology may increase the expected average years of schooling with more effective self-learning software and lower the cost of education in the future.
- any other relevant point

[Mark the FIRST TWO points only.]

} 2@
max: 4

(iii) Factors:

- Subsidise information technology/engineering department in the universities to boost the research and development.
- Provide tax allowances on education to increase the future productivity of labour.
- any other relevant point

[Mark the FIRST TWO points only.]

} 1@
max: 2

Candidates' Performance

Paper 1

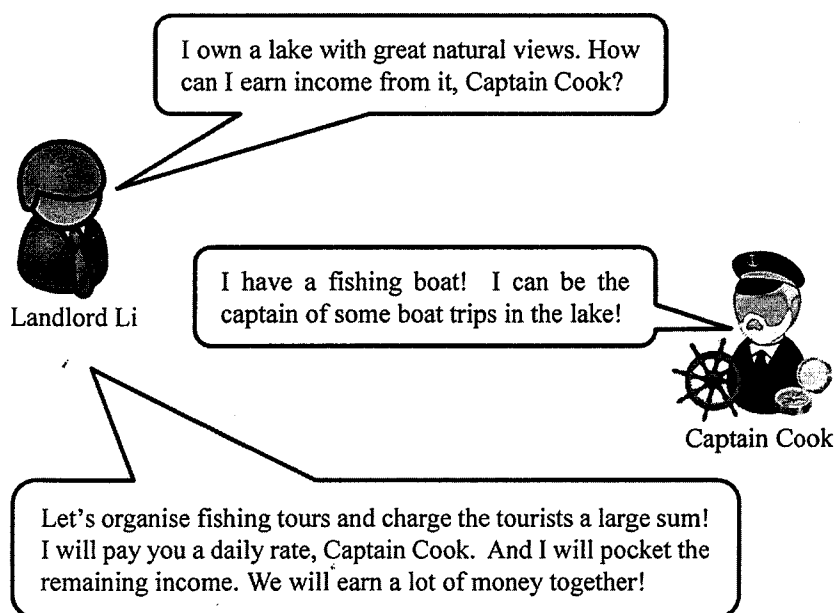
Candidates' overall performance was good. The mean score of this paper was 28 out of 45. The following questions deserve special attention.

Q.5 If a public limited company in Hong Kong gets listed on the stock exchange,

- A. it will face a higher profits tax rate. (7%)
- B. it can no longer keep its accounting information confidential. (42%)
- *C. its shareholders will enjoy higher liquidity of their assets. (40%)
- D. it will enjoy economies of scale. (11%)

Many candidates mistakenly picked option B. This suggests that they did not realise that a public limited company is required to submit its financial information to the Companies Registry, so that its financial information is accessible to the public whether it is listed on the stock exchange or not. The difference that 'listing' makes is that the company's shareholders can trade their stocks for cash on the stock market. Therefore, option C is the key.

Q.6 Refer to the following dialogue.



Given the return of land is rent, the return of labour is wage, the return of capital is interest while the return of entrepreneur is profit. The income earned by Captain Cook consists of _____ while the remaining income earned by Landlord Li consists of _____.

- A. wage.....profit (51%)
- B. interestrent and profit (3%)
- C. wage and interest.....rent (4%)
- *D. wage and interestrent and profit (42%)

This question is about factors of production and their corresponding incomes. A person may perform more than one role in the production process and may thus earn more than one kind of income. As a land owner and an entrepreneur, Landlord Li receives both rental and profit incomes. As a provider of both capital and labour services, Captain Cook earns interest and wage incomes respectively.

Q.25 Which of the following should be deducted from the gross domestic product of Hong Kong to calculate the gross national income of Hong Kong?

- *A. the fee paid by a Hong Kong football club to a Japanese football team for an exhibition match in Hong Kong (53%)
- B. the rent earned by a Hong Kong property developer from its residential property in Japan (10%)
- C. the capital gain earned by a Japanese insurance corporation from its financial investment in Hong Kong (27%)
- D. the dividend paid by a listed company in Japan to a Hong Kong resident (10%)

This question is about factor incomes from/to abroad. Most candidates correctly ruled out options B and D. Since factor incomes were earned by local residents from abroad in both cases, they should be added to (rather than deducted from) Hong Kong's GDP to calculate its GNI. Most capable candidates also managed to rule out option C, realising that capital gains are not considered as factor income. Option A is the best answer, as it describes a situation that might be conceived as a resident producing unit of Hong Kong employing which pays 'wages' to foreigners as a labour input in the production of the football match. Statistics shows that this question effectively discriminates able candidates from weaker ones.¹

Q.33 Which of the following will increase the actual banking multiplier?

- A. The public tends to hold more cash. (22%)
- *B. The use of credit card becomes more popular. (36%)
- C. The central bank increases the required reserve ratio. (18%)
- D. The interest rate of bank deposit decreases. (24%)

Most capable candidates were able to rule out options A, C, and D, which would all cause a reduction in actual deposits and reserves and tend to narrow the scope of credit creation. Option B implies that there is a drop in the public's willingness to hold cash, resulting in a larger quantity of deposits and reserves in the banking system. The banking multiplier would attain its maximal effect (i.e., actual deposits = maximum possible amount of deposits) when the following standard assumptions are satisfied—i.e., absence of cash leakage and of excess reserves held by banks. The further the actual economy deviates from these assumptions, the weaker the multiplier effect. If people use credit cards (or e-payments) to settle transactions more often, they need not hold too much cash in their pockets (i.e., less cash leakage). Instead, they would keep a larger amount of deposits in their banks and withdraw deposits less often. As banks expect such change in deposit-withdrawal behaviour, they need not hold a lot of reserves (i.e., less excess reserves); they can safely make more/bigger loans. Under this situation, when the standard assumptions are more or less satisfied, the actual banking multiplier would likely be close to its maximum possible level.²

¹ In standard textbooks, the classification of 'income paid abroad' focuses on the identity of the recipient (i.e., whether the sum of money is paid to a foreign, rather than domestic, resident) and the nature of income (i.e., whether the sum is viewed as factor income). Instead of classifying the fee mentioned in option A as such (i.e., as NIA), one may argue that this fee should better be treated as 'import of services'. In authentic contexts, the correct way to classify the fee is determined by the precise terms specified in the contract between the HK football club and the Japanese football team. Such complications in the accounting treatment of BOP involved in the actual arrangements of business activities are not the intention of this exam assessment. Students are not expected to look into such details in answering this question.

² It is not advisable to use a mathematical approach to tackle this question because it would be very difficult to calculate the change in the amount of actual deposits due to the change in cash leakage (i.e., cash-deposit ratio) and to the change in the banks' tendency to hold excess reserves (i.e., actual reserve ratio) in this context. As a side note, the manipulation of cash-deposit ratio in the analysis of credit creation is not required in the Curriculum.

Paper 2

Section A

Q. number	Performance in General
1	Satisfactory. The majority of candidates correctly pointed out that a free good is necessarily free-of-charge and supported their stand by some features of a free good. However, some candidates just recited the definition of free good without relating their arguments to the question, thus failing to obtain the full score for this question.
2(a)	Satisfactory. Some candidates failed to give a proper explanation for the type of expansion they named.
2(b)	Good. A few candidates listed some motives which did not suit the expansion in this case, e.g., increasing market share.
2(c)	Good. Instead of providing reasons for the declining average cost, some candidates simply listed additional motives behind the expansion. A few thought that the question was asking about increasing average cost and gave irrelevant answers.
3(a)	Excellent.
3(b)	Good. A few candidates did not read the question carefully and included 'increase in unemployment rate' in their answers.
3(c)	Good. Some candidates regarded an increase in welfare payment as a loss to society.
4(a)	Good. A few candidates had problems in rounding off the inflation figures.
4(b)	Satisfactory. Some candidates failed to explain the difference between CPI and GDP deflator from a single perspective. Instead, they merely stated one feature of CPI and another feature of GDP deflator without pointing out the difference.
4(c)	Satisfactory. Most candidates wrote the correct quantity equation, but only a few got full marks. The following lists some of the mistakes found in their answers: -failing to define the variables M, V, P, and Y; -failing to realise that classical QTM assumes that V and Y are constant; -failing to mention that inflation is a continuous and sustained increase in the price level; -mis-thinking that open market purchase of government bonds would result in a fall in the money supply.
5(a)	Good. Many candidates were able to point out that holding cash implies forgoing the option to earn interest income from various means of financial investment, such as bank deposits or bonds. Weaker candidates could only state the definition of opportunity cost without elaborating on the kind of investment return people have to give up while holding cash. Some simply put down the term 'interest' without explanation.
5(b)	Satisfactory. The majority of candidates were able to compare cash and the supermarket coupon from the perspective of 'functions of money' or 'features of good money'. A common problem was the inaccuracy or imprecision in using such terminology as general acceptability, divisibility, durability, medium of exchange, and store of value, resulting in vague or inconsistent answers.
6	Satisfactory. Some candidates failed to realise that the market adjustment operates via a leftward shift in the SRAS curve to restore output to its full-employment level. Instead, they shifted either AD or LRAS. A few candidates left out the reason behind the SRAS adjustment, i.e., a rise in production cost due to excess demand in the factor market.
7(a)	Good. In determining the direction of trade, a few candidates mistakenly compared the unit cost of cars with that of toys in a given country. Some missed out the unit in their calculations. Others did not state explicitly that country B enjoys comparative advantage in producing and exporting toys because it has a lower opportunity cost in producing toys than country A.

Q. number	Performance in General
7(b)	Satisfactory. Some candidates overlooked the keywords 'the cost is equally shared by both countries' in the question and thus got the wrong answer.
8(a)	Excellent.
8(b)(i)	Good. Many candidates demonstrated understanding of how the monetary base would change in the case of cash withdrawal.
8(b)(ii)	Good. Most candidates calculated the required reserve ratio correctly. However, some made the following mistakes: - neglecting the word 'change' and calculating only the new level of money supply; - failing to state verbally that there was a 'drop' in money supply OR failing to give a 'negative' figure as their numerical answer; - omitting the unit (i.e., \$ million) in their answers.

Section B

Q. number	Performance in General
9(a)	Good. The only problem was that a minority of candidates misspelled the term 'oligopoly' or created a term like 'oligopoly competition'.
9(b)	Satisfactory. Most candidates demonstrated understanding of the AS-AD model and correctly presented the case in a proper diagram. In their verbal explanation, many candidates were able to point out that a rise in tourist spending would lead to an increase in net export, but only some clearly stated that there was an increase in the export of services. Common errors included: -misconceiving an increase in tourist expenditure as a rise in consumption expenditure; -mistaking the expense on infrastructure development for an increase in government expenditure or an increase in labour supply; -failing to elaborate on how the case would lead to an increase in LRAS; -mislabelling the horizontal axis as price or \$.
9(c)	Satisfactory. Some candidates merely stated the advantages of issuing shares without mentioning in what ways it is better than obtaining bank loans as a means of raising capital. Some erroneously thought that issuing shares could help Ocean Park raise capital more easily or raise more capital than bank loans. Others mis-read the question and focused on the difference between shares and bonds.
9(d)	Fair. Many candidates were able to identify the source of discrepancy, but not everyone could provide a clear explanation. For instance, some candidates merely provided the terms 'inventory' or 'imports' without mentioning that these goods are not produced in the current period or not produced by domestic firms.
10(a)(i)	Satisfactory. Common mistakes included - missing the word 'balance' in their discussion of trade/current account balance; - mixing up the 'balance of goods and services' with 'balance of current account'.
10(a)(ii)	Satisfactory. Most candidates were able to point out that there are other sub-accounts within the current account. One common mistake was the confusion between 'current transfer' and 'transfer payment'.

Q. number	Performance in General
10(b)(i)(ii)	Satisfactory. Most candidates correctly stated one such condition, but a few candidates missed some important steps in their verbal explanation or made some errors in their diagrams. Common mistakes are listed below: - mis-matching perfectly elastic demand with a vertical demand curve OR perfectly inelastic supply with a horizontal supply curve; - failing to mention that the after-tax market price would stay unchanged, so that the consumers would bear no tariff burden; - failing to identify correctly the after-tax market price and/or the per-unit price the sellers can receive net-of-tax, and thus to indicate correctly the tariff revenue.
10(c)	Good. One common error was to wrongly include the tariff revenue in the after-tariff producers' surplus and mis-labelling the latter area as $bci0$.
10(d)	Satisfactory. The majority of candidates correctly suggested the advantage and disadvantage from the viewpoint of the US government. However, some erroneously thought that tariff would lead to a rise in import prices while quota would not, OR that tariff might lead to retaliation from other countries while quota would not.
11(a)	Good. A few candidates simply provided a sentence or two without saying whether it was an advantage or disadvantage.
11(b)	Good. Most candidates were able to illustrate the traffic-jam situation using a supply-demand diagram, but only some got full credits. Common mistakes/omissions included: - failing to state that 'the price is set below its equilibrium (market-clearing) level' in their verbal elaboration; - drawing an upward-sloping, rather than vertical, supply curve; - stating imprecisely that shortage corresponds to the situation where demand is bigger than supply (rather than 'the quantity demanded is bigger than the quantity supplied at the given price/toll'); - referring to a case of an increase in demand (not mentioned in the given information) as the cause of shortage in their diagrams.
11(c)(i)	Fair. Some candidates did not mention that other drivers using the tunnel are adversely affected, or that they are not compensated for these adverse effects. Others failed to use the information from source C to support their answers, e.g., they cited air pollution as an example of external cost.
11(c)(ii)	Fair. Only some of the candidates were able to pinpoint the gist of the inefficiency in this case—i.e., $MSC > MSB$ and over-use of the tunnel. Instead, some wrongly explained inefficiency by comparing marginal private cost with marginal private benefit, or claimed that $MSC > MSB$ would result in under-production.
11(d)	Excellent.
11(e)	Satisfactory. The following are some general comments. - Many candidates were able to organise their arguments on behalf of different stakeholders in separate paragraphs, and present them in an orderly way. - A few candidates provided a list of the effects of both proposals, which was not required. - While considering the preference of a given stakeholder, some candidates sometimes included discussion of some other stakeholders, to the detriment of the clarity of their answers. - Some weaker candidates just copied data from the sources without analysing them before concluding which proposal is preferred by which stakeholder. Comments on sub-questions are listed below: <u>For the owners of the WHC company</u> Some candidates failed to realise that WHC is a private firm so the original price it sets should be profit-maximising. Others failed to use more precise terms, such as a change in the demand for WHC service or competitive demand, in their elaborations. Instead, they described the situation loosely as a change in the usage of the 2 tunnels upon toll adjustments under the 2 proposals.

Q. number	Performance in General
	<u>For income inequality</u> Quite a number of candidates were able to provide proper reasons to support their choice between Proposal I and Proposal II. But some of their explanations were incomprehensive. For example, some candidates argued for Proposal I on the grounds that the toll raise would only affect the rich, without mentioning that most rich people drive private cars while the poor commute by public transport. Others argued for Proposal II on the ground that the toll cut would relieve the poor's burden, without mentioning that the percentage drop in toll (as a regressive tax) for the low-income group is bigger than that for the high-income group. <u>For government revenue and/or expenditure</u> Some candidates were able to give concise and precise explanations to demonstrate their in-depth understanding of public finance. In discussing the effects of externality or the difficulty in negotiation with WHC about the toll adjustment, however, some merely referred to the information in Source C or Source F without addressing the gist of the question. <u>Other stakeholders</u> Most candidates tried hard to identify a group of stakeholders and to deduce their stand as a group by examining the effects of the two proposals on such a group. One common problem was that they simply named their chosen group vaguely as 'people,' or 'citizens,' or 'drivers,' making it difficult to analyse precisely the effects of the two proposals on different members of the group separately and then to aggregate the effects across these members to reach a conclusion for the group as a whole. For those who picked a better-defined group of stakeholders (e.g., taxi drivers, or environmentalists), their discussion sometimes neglected the fact that both proposals can reduce the traffic jam, so alleviation of traffic congestion cannot be used to support one proposal against another.
11(f)	Satisfactory. A few candidates did not read the question carefully, and stated taxation on cross harbour tunnels or toll adjustment as their answers.

Section C

Q. number	Popularity	Performance in General
12(a)	58%	Satisfactory. Some candidates wrongly drew an upward-sloping marginal cost curve. Others did not state the profit-maximising condition in their verbal elaboration, or simply stated 'marginal benefit = marginal cost' as the condition.
12(b)		Poor. Many candidates did not realise that the imposition of a lump-sum tax would not affect marginal cost or marginal revenue.
12(c)		Good. Most candidates were able to apply their knowledge of price discrimination in answering this question. Many of them correctly identified the higher cost of electricity production in the rural area as a proper explanation for (non-discriminatory) differential pricing.
12(d)(i)		Excellent.
12(d)(ii)		Satisfactory. Some candidates wrongly stated the harm such anti-competitive behaviour would inflict upon the consumers, instead of ABC Company. Others wrongly conceived that the company had violated the law and would thus be punished.
13(a)	42%	Good. Some candidates overlooked the given information and calculated the opportunity cost of producing 1X in country A as 1Y (=100/100).
13(b)		Satisfactory. Some candidates did not realise that country A would not specialise completely in producing one single good. Others had difficulty in explaining how they could find the production point of country A in part (ii).
13(c)(i)		Satisfactory. A few candidates misread the question and thought that the productivity of both Good X and Good Y doubled simultaneously. Others did not support their answer with a proper calculation of the new opportunity cost.
13(c)(ii)		Fair. One common mistake was to use GNI, instead of per capita GNI, to explain the AI-effect on economic development. Overlooking the difference between economic growth and economic development, some candidates focused their discussion narrowly on per capita GNI and ignored the other two aspects of HDI (i.e., education and health).
13(c)(iii)		Good. Candidates should be reminded that, while increases in the quantity/quality of capital goods or in the quality of labour would help boost economic growth via an increase in GNI, any factor contributing to an increase in the population only may not lead to an increase in per capita GNI (despite its positive effect on GNI).

General comments and recommendations

1. Some candidates failed to demonstrate a thorough understanding of basic economic concepts and principles, especially when applying them to solve hypothetical or real-life problems. Candidates should strengthen their analytical skills instead of simply regurgitating concepts and theories.
2. Some candidates lacked skills to extract information from tables and graphs as well as to draw well-labelled diagrams. Candidates should pay special attention to the headings, labels, and axes as they are essential to our understanding of the data presented therein.
3. Some candidates had difficulty in presenting their answers in a clear and precise way. Candidates should equip themselves with better language abilities as well as graphical skills, and make sure that their handwriting is legible.
4. Some candidates did not read the questions carefully. Candidates should devote special attention to the assumptions and conditions laid down in the questions to avoid giving irrelevant answers.
5. Some candidates did not seem to be well-informed about current economic issues in the real world. Candidates should pay more attention to economic news.

Acknowledgements

The following material has been used in question papers in this volume:

Mingpao	租金 10 年倍增德國柏林通過明年起凍租 5 年 (Chinese version only) https://news.mingpao.com/ins/%e5%9c%8b%e9%9a%9b/article/20190618/s00005/1560865099102/%e7%a7%9f%e9%87%9110%e5%b9%b4%e5%80%8d%e5%a2%9e-%e5%be%b7%e5%9c%8b%e6%9f%8f%e6%9e%97%e9%80%9a%e9%81%8e%e6%98%8e%e5%b9%b4%e8%b5%b7%e5%87%8d%e7%a7%9f5%e5%b9%b4
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South China Morning Post	Pay gap: Hong Kong women earn more than men before 20 but only half when they turn 60 https://www.scmp.com/business/companies/article/2149404/pay-gap-hong-kong-women-earn-more-men-20-only-half-when-they-turn
Census and Statistics Department	Labour: Latest Statistics https://www.censtatd.gov.hk/hkstat/sub/so30.jsp
Transport Department	Table 2.2: Existing Tunnel Traffic Flows, Queues and Crossing Time https://www.td.gov.hk/filemanager/en/publication/toll%20rationalisation%20study%20-%20final%20report.pdf
Transport Department	5.3 Cross Harbour Vehicular Trips https://www.td.gov.hk/filemanager/en/content_4953/annual%20traffic%20census%202018.pdf

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